

Spondyloepimetaphyseal Dysplasia, Faden-Alkuraya Type (SEMDFA): A Comprehensive Disease Characteristics Report

Disease: Spondyloepimetaphyseal Dysplasia, Faden-Alkuraya Type (SEMDFA) **MONDO ID:** MONDO:0014748 | **OMIM:** #616723 | **Orphanet:** ORPHA:457395 **Causal Gene:** *RSPRY1* (Ring Finger and SPRY Domain Containing 1), 16q13 **Category:** Mendelian (autosomal recessive)

Summary

Spondyloepimetaphyseal dysplasia, Faden-Alkuraya type (SEMDFA) is an ultra-rare, autosomal-recessive skeletal dysplasia first delineated in 2015 and caused by biallelic loss-of-function variants in *RSPRY1*, a gene on chromosome 16q13 encoding a 576-amino-acid secreted protein that contains an N-terminal RING (Really Interesting New Gene) domain and a C-terminal B30.2/SPRY (PRY-SPRY) domain. The disease is defined clinically by a recognizable constellation of **progressive spondyloepimetaphyseal dysplasia, disproportionate short stature, facial dysmorphism, short (fourth) metatarsals and cono-brachydactyly, craniosynostosis, and variable intellectual disability**. Consanguinity is the principal risk factor, and all reported families to date derive from consanguineous or founder backgrounds (Saudi Arabian, Turkish, Peruvian, Indian).

Mechanistically, the most important recent advance is the demonstration that *RSPRY1* deficiency causes **constitutive, SMAD3-dependent activation of TGF- β signaling** with dysregulated extracellular-matrix (ECM) dynamics. Transcriptomic profiling of patient fibroblasts showed significant enrichment of TGF- β and ECM pathways; *RSPRY1*-knockout fibroblasts exhibited enhanced motility that was abolished by concurrent *SMAD3* knockout, and *RSPRY1*-deficient cells responded poorly to exogenous TGF- β — the signature of a pathway

already maximally (constitutively) activated. This places SEMDFA within the broader family of TGF- β -dysregulated skeletal dysplasias and nominates TGF- β /SMAD3 inhibition as a rational, though still entirely experimental, therapeutic hypothesis.

RSPRY1 is one of the more constrained genes in the human genome (gnomAD pLI $\approx$ 1.0; LOEUF 0.45; missense Z = 4.05), consistent with strong purifying selection against both loss-of-function and missense variation, and with the severe developmental phenotype produced by complete loss of function. The variant spectrum spans frameshift, nonsense, canonical splice-site, and missense alleles, plus large 16q contiguous-gene deletions that encompass *RSPRY1* and produce overlapping skeletal features. There is currently no disease-specific therapy; management is supportive (orthopedic correction of coxa vara/genu valgum, developmental support, monitoring for craniosynostosis) and prevention relies on genetic counseling with carrier and prenatal/preimplantation testing in at-risk families.

Key Findings

Finding 1 — SEMDFA is an autosomal-recessive dysplasia caused by biallelic loss-of-function *RSPRY1* variants

SEMDFA was established as a distinct, recognizable Mendelian entity by Faden and colleagues in 2015 using combined autozygome/exome analysis. In a consanguineous Saudi family with four affected siblings, they identified a **homozygous frameshift mutation in *RSPRY1* resulting in nonsense-mediated mRNA decay (NMD)** — i.e., a true null allele — and confirmed the locus in an independent simplex Peruvian case carrying a homozygous missense variant. The original report noted: "*Combined autozygome/exome analysis identified a homozygous frameshift mutation in *RSPRY1* with resulting nonsense-mediated decay*" ([PMID: 26365341](#)).

The recessive loss-of-function mechanism has been reproduced across multiple independent families and variant classes:

Study	Family origin	Variant(s)	Type	Patients
Faden et al. 2015 (PMID: 26365341)	Saudi (consanguineous)	Homozygous frameshift → NMD	Frameshift/ null	4 siblings
Faden et al. 2015	Peruvian (simplex)	Homozygous missense	Missense	1
Simsek-Kiper et al. 2018 (PMID: 30063090)	Two families	c.377delT p.Ile126fs*; c.516+2T>A (splice)	Frameshift; splice	5
Singh et al. 2024 (PMID: 38562122)	Two sisters	c.1652G>A p. (Cys551Tyr)	Missense	2

Simsek-Kiper et al. reported: "*Whole exome sequencing revealed a novel homozygous [c.377delT] [p.Ile126fs*] frameshift mutation at exon 2 in one family, while Sanger sequencing revealed a novel homozygous splice site mutation [c.516+2T>A] at exon 4/intron 4 border of RSPRY1 in the other family*" ([PMID: 30063090](#)). The convergence of frameshift, nonsense, splice, and missense variants — all homozygous, all in consanguineous or founder settings — firmly supports a **loss-of-function, autosomal-recessive** disease model. OMIM assigns the disease number #616723; *RSPRY1* (HGNC:29420; NCBI Gene 89970) maps to chromosome **16q13**.

Finding 2 — Core clinical phenotype: progressive spondyloepimetaphyseal dysplasia with short stature, facial dysmorphism, cono-brachydactyly, and intellectual disability

The phenotype is remarkably consistent across all reported cohorts. The original description defined the syndrome as "*comprising progressive spondyloepimetaphyseal dysplasia, short stature, facial dysmorphism, short fourth metatarsals, and intellectual disability*" ([PMID: 26365341](#)).

Simsek-Kiper et al. (n = 5) provided the most detailed radiographic delineation, defining the skeletal hallmarks as: "*(a) mild spondylar dysplasia, (b) epimetaphyseal dysplasia of the long bones associated with coxa vara and genu valgum, (c) brachymesophalangy with cone-shaped epiphyses, and (d) craniosynostosis*" ([PMID: 30063090](#)). They additionally documented pes planus, prominent heels, pectus deformity, and — notably — a cemento-ossifying fibrous lesion of the maxilla in one patient.

Singh et al. (2024) further expanded the spectrum, reporting **joint dislocation as a novel clinical feature**: "*We observed joint dislocation as a novel clinical feature of this condition*" (PMID: [38562122](#)), alongside vertebral defects, small epiphyses, and metaphyseal cupping/fraying.

Suggested HPO-annotated phenotype profile:

Phenotype	HPO term	Onset / character	Frequency
Progressive spondyloepimetaphyseal dysplasia	HP:0002762 (SEMD spectrum)	Congenital→childhood, progressive	Universal
Short stature (often disproportionate)	HP:0004322	Childhood, progressive	Universal
Facial dysmorphism	HP:0001999	Congenital	Universal
Intellectual disability (variable)	HP:0001249	Childhood	Most (variable)
Short (fourth) metatarsals	HP:0004689	Childhood	Common
Brachydactyly / brachymesophalangy	HP:0001156 / HP:0005831	Childhood	Common
Cone-shaped epiphyses	HP:0010579	Childhood	Common
Craniosynostosis	HP:0001363	Congenital/early	Reported cohort
Coxa vara	HP:0002812	Progressive	Common
Genu valgum	HP:0002857	Childhood	Common
Mild spondylar/vertebral dysplasia	HP:0000925	Progressive	Common
Metaphyseal cupping/fraying	HP:0003023 / HP:0003021	Childhood	Reported (Singh 2024)
Pectus / pes planus	HP:0000766 / HP:0001763	Childhood	Reported
Joint dislocation	HP:0001373	Childhood	Novel (Singh 2024)

Severity is best characterized as **moderate-to-severe and progressive** for the skeletal component, with **variable** intellectual disability. Cumulative published patients number ~11 across 5 families, so robust per-phenotype frequency percentages are not available; the features above are qualitatively consistent ("core") versus variable/expanding.

Finding 3 — RSPRY1 deficiency dysregulates TGF- β /SMAD3 signaling and ECM dynamics (the molecular mechanism)

The pivotal mechanistic study is Imren et al. 2025 (*Int J Mol Sci*), which performed transcriptomic analysis of fibroblasts from patients with homozygous *RSPRY1* mutations. They reported: *"Transcriptome analysis of fibroblasts from patients with homozygous RSPRY1 mutations showed there was significant enrichment of transforming growth factor beta (TGF- β) signaling and ECM-related pathways"* (PMID: 39940902).

Two functional experiments establish the causal, SMAD3-dependent nature of the mechanism:

1. **Epistasis / motility rescue:** *"RSPRY1 knockout fibroblasts exhibited enhanced motility, a phenotype that was abrogated in RSPRY1 + SMAD3 double knockout fibroblasts, highlighting the SMAD3-dependence of RSPRY1's effects"* (PMID: 39940902). The rescue of the cellular phenotype by removing SMAD3 places SMAD3 genetically downstream of RSPRY1 loss.
2. **Constitutive activation:** *"The observed limited response to exogenous TGF- β in RSPRY1-deficient cells indicated that there was constitutive pathway activation"* (PMID: 39940902). A blunted response to added ligand is the hallmark of a pathway already operating near its ceiling.

Together these define the disease as one of **de-repressed / constitutive TGF- β –SMAD3 signaling**, consistent with the well-established role of TGF- β superfamily signaling in chondrocyte proliferation/differentiation and endochondral ossification, and consistent with the observed epimetaphyseal (growth-plate) pathology.

Suggested GO / pathway terms: GO:0007179 (transforming growth factor beta receptor signaling pathway); GO:0060395 (SMAD protein signal transduction); GO:0030198 (extracellular matrix organization); GO:0001958 (endochondral ossification); GO:0002062 (chondrocyte differentiation); Reactome R-HSA-170834 (Signaling by TGF- β Receptor Complex).

Finding 4 — Genomic context (16q12.2–q21), contiguous-gene deletions, and skeletal-muscle expression

RSPRY1 lies within the 16q12.2–q21 region. Yamamoto et al. (2016) described a patient with a 16q12.2q21 interstitial deletion encompassing *RSPRY1* who presented with developmental delay, epilepsy, short stature, and distinctive features, and proposed: *"Because RSPRY1 was been discovered as the cause of progressive skeletal dysplasia, a loss of this gene might explain*

the skeletal defects observed in the patient" (PMID: 27230627). This supports a contribution of *RSPRY1* to the skeletal component of larger copy-number syndromes, while the full recessive disease requires biallelic loss.

At the expression level, Waddell et al. (2016) showed that *Rspry1* is expressed in skeletal muscle and is transcriptionally regulated during muscle wasting: *"Nip30 and Rspry1 were transcriptionally induced in response to neurogenic muscle wasting in mice and were also found to be expressed endogenously at the RNA and protein level in C2C12 mouse muscle cells"* (PMID: 26497270). These two genes share a regulatory region with a conserved MyoD1-bound E-box enhancer.

The tissue relevance to bone is anchored by the original report's immunolocalization data: *"we detect strong RSPRY1 protein localization in murine embryonic osteoblasts and periosteal cells during primary endochondral ossification, consistent with a role in bone development"* (PMID: 26365341).

Finally, Jacob et al. (2025), a cohort of 248 Indians with skeletal dysplasia (73.6% diagnostic yield; 60% autosomal recessive; 35% consanguinity), expanded the genotype/phenotype spectrum, listing *RSPRY1* among rarely reported conditions confirmed in their cohort: *"We expand the phenotypic and genotypic spectrum of rarely reported conditions (RAB33B, TRIP11, NEPRO, RPL13, COL27A1, PTHR1, EXOC6B, PRKACA, FUZ and RSPRY1)"* (PMID: 39706863).

Finding 5 — Authoritative identifiers and RSPRY1 protein annotation

Cross-referenced ontology identifiers: **MONDO:0014748** ("progressive spondyloepimetaphyseal dysplasia-short stature-short fourth metatarsals-intellectual disability syndrome"); **OMIM #616723**; **Orphanet ORPHA:457395**; **GARD 0017808**; **UMLS C5568882**; **MedGen 1800305**. Common synonyms: SEMDFA; "spondyloepimetaphyseal dysplasia, Faden-Alkuraya type"; "RSPRY1-associated SEMD with cono-brachydactyly and craniosynostosis."

The RSPRY1 protein corresponds to **UniProt Q96DX4 (RSPRY_HUMAN)**, is **576 amino acids** long, is annotated with a **Secreted** subcellular localization, and contains a **B30.2/SPRY domain (≈ residues 300–483)** and an **N-terminal RING domain**. The original report described it as *"a hypothetical RING and SPRY domain-containing protein of unknown physiological function"* (PMID: 26365341). The RING + PRY/SPRY architecture is shared with the TRIM (tripartite-motif) family of E3 ubiquitin ligases and substrate-recognition proteins, in which PRY-SPRY is the domain "most frequently used for substrate recognition" (PMID: 33092958) — a plausible structural basis for RSPRY1 acting as a ubiquitin-ligase/adaptor that normally restrains TGF-β signaling.

Finding 6 — Expanded pathogenic variant spectrum from ClinVar

ClinVar (reference transcript NM_133368.3) contains ~244 *RSPRY1* variant records, of which ~31 are classified pathogenic/likely pathogenic. Beyond the published case-report alleles, curated pathogenic/likely-pathogenic small variants include:

Variant (cDNA)	Protein	Class	Classification
c.109del	p.Ala37fs	Frameshift	Pathogenic
c.358G>T	p.Glu120Ter	Nonsense	Pathogenic
c.1422T>A	p.Cys474Ter	Nonsense	Pathogenic
c.1279dup	p.Thr427fs	Frameshift	Pathogenic
c.121G>T	p.Gly41Cys	Missense	Pathogenic/LP
c.901+3_901+15del / c.901_901+15del	—	Splice-region	Conflicting/uncertain
c.1652G>A	p.(Cys551Tyr)	Missense	Pathogenic (Singh 2024)

The published missense allele was confirmed clinically: "*harbored a homozygous missense variant c.1652G>A;p.(Cys551Tyr) in the RSPRY1 gene*" ([PMID: 38562122](#)). Many additional pathogenic ClinVar entries are large 16q contiguous-gene copy-number gains/losses (e.g., 16q13-21 deletions) spanning *RSPRY1*. Nearly all point variants are **germline** and homozygous; no somatic disease association is reported.

Finding 7 — *RSPRY1* is highly constrained in the general population

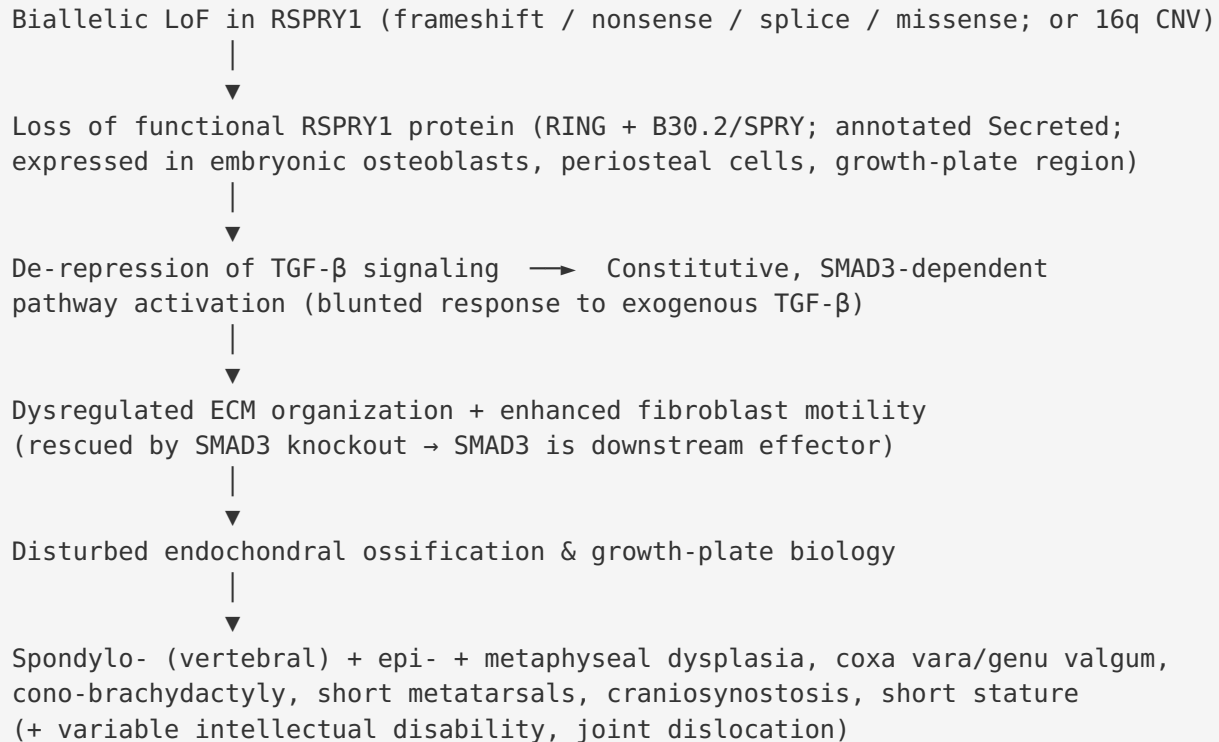
gnomAD constraint metrics (GRCh38; Ensembl ENSG00000159579) indicate strong purifying selection:

Metric	Value	Interpretation
pLI	0.9995	Extremely LoF-intolerant
Observed/expected LoF (o/e)	0.315 (90% CI 0.226–0.446)	~68% depletion of LoF
LOEUF	0.446	High constraint
LoF Z	4.97	Strong LoF depletion
Missense Z	4.05	Significant missense depletion
Synonymous Z	0.95	Neutral (as expected)

The strong depletion of both LoF and missense variation is consistent with a gene whose complete biallelic loss produces a severe developmental disorder, supports the pathogenicity of the reported missense alleles, and explains the near-absence of biallelic LoF in the general population. Heterozygous carriers are clinically unaffected (recessive disease); the constraint reflects gene essentiality/selection at the population level.

Mechanistic Model / Interpretation

The findings assemble into a coherent causal chain from genotype to skeletal phenotype:



Upstream vs downstream: The upstream trigger is loss of RSPRY1 function. The proximate downstream event is constitutive TGF-β/SMAD3 activation; SMAD3 sits genetically downstream (its removal rescues the RSPRY1-null cellular phenotype). ECM dysregulation and abnormal cell motility are further downstream cellular readouts, and disturbed endochondral ossification is the tissue-level consequence that yields the clinical skeletal dysplasia.

Cell types and structures involved: osteoblasts (CL:0000062), chondrocytes (CL:0000138), preosteoblasts (CL:0007010), periosteal/perichondrial cells, and fibroblasts (CL:0000057) — the last being the experimental system in which the mechanism was established. Anatomically the disease targets the **growth plate / metaphysis and epiphysis of long bones (UBERON:0006588 epiphyseal plate; UBERON:0004421 metaphysis; UBERON:0006755 epiphysis), the vertebral column (UBERON:0001130), the cranial sutures (UBERON:0007842) via craniosynostosis, and the hands/feet** (short metatarsals, brachydactyly). Subcellular compartments implicated include the extracellular/secreted space (GO:0005576; GO:0031012 extracellular matrix) and, given the RING/SPRY architecture, potentially the ubiquitin–proteasome machinery (GO:0004842 ubiquitin-protein transferase activity) — the latter unconfirmed for RSPRY1.

Why the RING/SPRY architecture matters: RSPRY1 shares its domain layout with TRIM-family E3 ubiquitin ligases, in which the PRY-SPRY (B30.2) domain is the principal substrate-recognition module (PMID: 33092958). A parsimonious hypothesis — not yet directly demonstrated for RSPRY1 — is that RSPRY1 normally ubiquitinates or otherwise restrains a component of the TGF- β /SMAD3 cascade; its loss removes that brake, producing constitutive signaling. Note that the UniProt "Secreted" annotation sits in apparent tension with a classic intracellular E3-ligase role; resolving this is a key open question.

Section-by-Section Disease Characteristics

1. Disease Information

SEMDFA is an ultra-rare autosomal-recessive skeletal dysplasia characterized by progressive spondyloepimetaphyseal dysplasia, short stature, facial dysmorphism, cono-brachydactyly/short metatarsals, craniosynostosis, and variable intellectual disability. It was first delineated in 2015 in a consanguineous Saudi family and named after the lead investigators Faden and Alkuraya. Identifiers: MONDO:0014748; OMIM #616723; ORPHA:457395; GARD 0017808; UMLS C5568882; MedGen 1800305. ICD-10 maps only to the non-specific Q77.8 (osteochondrodysplasia); no specific MeSH term exists (indexed under "Osteochondrodysplasias"). Synonyms: SEMDFA; spondyloepimetaphyseal dysplasia, Faden-Alkuraya type; "RSPRY1-associated SEMD with cono-brachydactyly and craniosynostosis." Information is derived from **individual patient case reports** (~11 patients aggregated from 5 families), not from population-level EHR or registry resources.

2. Etiology

Primary cause: biallelic loss-of-function variants in *RSPRY1* (purely genetic; Mendelian recessive). **Genetic risk factors:** homozygous/compound-heterozygous LoF *RSPRY1* alleles; the gene is highly constrained (pLI $\approx$ 1.0). **Environmental risk factors:** none identified; the dominant non-genetic contributor to risk is **consanguinity** (family structure), which increases homozygosity for founder alleles. **Protective factors / gene-environment interactions:** none established; not applicable to a fully penetrant recessive Mendelian disorder.

3. Phenotypes

See Finding 2 table for the full HPO-annotated phenotype list, onset, and severity. Phenotype types span physical/skeletal manifestations (dysplasia, brachydactyly, coxa vara, genu valgum, craniosynostosis), clinical signs (short stature, dysmorphism, joint dislocation), and neurodevelopmental features (intellectual disability). Onset is congenital-to-childhood with a **progressive** skeletal course. Quality-of-life impact is driven by short stature, orthopedic deformity/mobility limitation, and developmental/cognitive impairment; formal QoL instrument data (EQ-5D, SF-36, PROMIS) are not available for this ultra-rare disease.

4. Genetic/Molecular Information

Causal gene: *RSPRY1* (HGNC:29420; NCBI Gene 89970; Ensembl ENSG00000159579; 16q13; reference transcript NM_133368.3). **Variant classes:** frameshift, nonsense, canonical splice-site, missense, and large 16q CNVs (Findings 1, 4, 6). **Functional consequence:** loss of function (NMD for truncating alleles; splice disruption; SPRY-domain missense likely disrupts substrate binding/folding). **Population frequency:** pathogenic alleles are extremely rare/private to consanguineous families; the gene is strongly depleted of LoF/missense variation in gnomAD (Finding 7). **Origin:** germline, biallelic. **Modifier genes / epigenetics:** none established; no methylation/histone data. **Chromosomal abnormalities:** contiguous-gene 16q12.2–q21 deletions represent the main structural abnormality involving the locus (Finding 4).

5. Environmental Information

Not applicable as a cause. No toxic, infectious, lifestyle, or occupational agents are implicated. Consanguinity is a population/social risk factor for homozygosity, not an environmental disease cause.

6. Mechanism / Pathophysiology

Constitutive, SMAD3-dependent TGF- β pathway activation with ECM dysregulation, downstream of *RSPRY1* loss, disturbing endochondral ossification (Finding 3; Mechanistic Model above). Molecular profiling to date is transcriptomic (patient fibroblasts, bulk RNA-seq) plus functional CRISPR-knockout fibroblast models. Proteomic, metabolomic, single-cell, and spatial data are not yet available. No enzyme deficiency, immune dysregulation, or metabolic derangement is reported. Tissue damage is developmental (abnormal growth-plate/epimetaphyseal ossification) rather than degenerative or inflammatory. Suggested terms: GO:0007179, GO:0060395, GO:0030198, GO:0001958; Reactome R-HSA-170834.

7. Anatomical Structures Affected

Organ/system: skeletal system (primary) — vertebrae, long-bone epiphyses/metaphyses, cranium (sutures), hands/feet; nervous system (intellectual disability); connective tissue/ECM; joints (dislocation). **Cells:** osteoblasts (CL:0000062), chondrocytes (CL:0000138), fibroblasts (CL:0000057), periosteal cells. **UBERON:** epiphyseal plate (UBERON:0006588), metaphysis (UBERON:0004421), epiphysis (UBERON:0006755), vertebral column (UBERON:0001130), cranial suture (UBERON:0007842). **Subcellular:** extracellular space/ECM (GO:0005576, GO:0031012), consistent with the secreted annotation. **Lateralization:** generalized, bilateral, symmetric.

8. Temporal Development

Congenital-to-early-childhood onset; **chronic, insidious** pattern. The skeletal disease is explicitly **progressive** (spondylar changes, coxa vara, deformities worsen during growth), whereas neurocognitive impairment is static/developmental. Lifelong disorder; no spontaneous remission; no genetic anticipation (recessive, non-repeat-expansion). The critical window for orthopedic intervention corresponds to growth-plate-active childhood/adolescence.

9. Inheritance and Population

Inheritance: autosomal recessive — *"a clinically recognizable autosomal-recessive disorder in four affected siblings from a consanguineous Saudi family"* (PMID: 26365341). **Penetrance:** effectively complete for biallelic LoF. **Expressivity:** variable (intellectual-disability degree; head circumference — microcephaly in the original family vs normocephaly in the 2018 cohort; craniosynostosis). **Epidemiology:** ultra-rare; prevalence/incidence unknown (~11 published patients; Orphanet <1/1,000,000). **Founder/consanguinity:** central — all families consanguineous; homozygosity via autozygosity; reported in Saudi, Turkish, Peruvian, and Indian families. **Carrier frequency:** unknown, expected very low. **Sex ratio:** no sex bias (autosomal recessive). **Recurrence risk:** 25% per pregnancy for two carrier parents.

10. Diagnostics

Diagnosis rests on a **characteristic skeletal survey** (mild spondylar dysplasia; epimetaphyseal dysplasia with coxa vara/genu valgum; cone-shaped epiphyses/brachymesophalangy; metaphyseal cupping/fraying; short metatarsals; craniosynostosis) combined with **molecular confirmation of biallelic *RSPRY1* variants**. Genetic testing: **whole-exome sequencing** was the diagnostic modality in all discovery reports; WGS and skeletal-dysplasia gene panels including *RSPRY1* are alternatives; single-gene Sanger sequencing confirms/segregates.

Autozygosity mapping is powerful in consanguineous families; GeneMatcher-style matchmaking was pivotal — *"Using a gene-centric 'matchmaking' system, we were able to identify a Peruvian simplex case subject"* (PMID: 26365341). **Chromosomal microarray** detects large 16q contiguous-gene deletions that sequencing panels may miss (PMID: 27230627). Karyotype/FISH/mtDNA/repeat-expansion testing are not relevant. No specific biochemical biomarker or newborn-screening test exists. Differential diagnoses include other spondyloepimetaphyseal/spondylometaphyseal dysplasias, Dyggve-Melchior-Clausen dysplasia, and cone-shaped-epiphysis syndromes; *RSPRY1* genotyping is discriminating.

11. Outcome / Prognosis

No reports indicate reduced life expectancy; the disorder is a **chronic, non-lethal skeletal dysplasia** compatible with survival to adulthood. Morbidity is dominated by short stature, progressive orthopedic deformity (coxa vara, genu valgum, joint dislocation) and mobility limitation, plus variable cognitive disability; craniosynostosis may require neurosurgical attention. Skeletal changes are structural and not reversible. Prognostic factors are inferred (deformity severity, degree of intellectual disability); no validated prognostic biomarkers exist.

12. Treatment

There is **no disease-specific or disease-modifying therapy**. Management is **supportive and multidisciplinary**: orthopedic surgical correction of coxa vara/genu valgum and joint dislocations (NCIT: orthopedic surgical procedure), craniosynostosis repair where indicated, physical/occupational therapy and rehabilitation, and developmental/educational support for intellectual disability. Mechanistically, the constitutive SMAD3-dependent TGF- β activation nominates **TGF- β /SMAD3 pathway inhibition as a candidate therapeutic hypothesis** (PMID: 39940902) — entirely experimental (candidate agents such as ALK5/SMAD3 inhibitors or losartan-type TGF- β modulation used in other connective-tissue disorders), with no preclinical or clinical validation in SEMDFA and no registered trials. No pharmacogenomic considerations are established.

13. Prevention

Primary prevention is via **genetic counseling** in consanguineous/at-risk families, **carrier testing**, and reproductive options including **prenatal diagnosis and preimplantation genetic testing (PGT-M)** once the familial *RSPRY1* variant is known. Cascade testing identifies at-risk

relatives. Secondary/tertiary prevention focuses on early orthopedic and developmental intervention to limit complications. No population-based screening, immunization, or environmental interventions apply.

14. Other Species / Natural Disease

No naturally occurring SEMDFA-equivalent disease is documented in companion animals or wildlife (no OMIA entry identified). Orthologous genes exist across vertebrates (mouse *Rspry1*; NCBI Taxon *Homo sapiens* 9606, *Mus musculus* 10090); the gene is expressed in mouse skeletal muscle and C2C12 cells (PMID: 26497270), and murine embryonic bone was used to localize the protein (PMID: 26365341). No zoonotic or cross-species transmission relevance (Mendelian, non-infectious).

15. Model Organisms

No dedicated in vivo *Rspry1* knockout disease model has been reported. Available systems are **cellular/in vitro**: patient-derived dermal fibroblasts with homozygous *RSPRY1* mutations and engineered **RSPRY1-knockout ($\pm$ SMAD3 double-knockout) fibroblasts** used to establish the TGF- β /SMAD3 mechanism (PMID: 39940902), plus mouse **C2C12 muscle cells** for expression studies (PMID: 26497270). Recommended future models: an *Rspry1* knockout/knock-in mouse (IMPC/KOMP), zebrafish *rspry1* CRISPR knockdown for skeletal phenotyping, and patient-derived iPSC $\rightarrow$ osteoblast/chondrocyte or organoid systems. Phenotype recapitulation is **unknown** — a major research gap.

Evidence Base

PMID	Title (abbrev.)	Contribution	Evidence type
26365341	<i>Identification of a Recognizable Progressive Skeletal Dysplasia Caused by RSPRY1 Mutations</i>	Founding report: gene discovery, NMD LoF mechanism, core phenotype, osteoblast localization	Human clinical + in vitro
30063090	<i>Further delineation of SEMDFA...</i>	Radiographic hallmarks; new frameshift + splice alleles; craniosynostosis/cono-brachydactyly	Human clinical
38562122	<i>Two sisters with RSPRY1-related SEMD</i>	Homozygous missense c.1652G>A p.(Cys551Tyr); novel joint dislocation	Human clinical
39940902	<i>Unraveling the Role of RSPRY1 in TGF-β Pathway Dysregulation</i>	Core mechanism: constitutive SMAD3-dependent TGF- β activation; ECM dysregulation	In vitro / patient cells
27230627	<i>A 16q12.2q21 deletion...</i>	Contiguous-gene deletion including <i>RSPRY1</i> explains skeletal defects	Human clinical
26497270	<i>NIP30 and RSPRY1 in skeletal muscle</i>	<i>Rspry1</i> muscle expression/regulation; MyoD1 E-box enhancer	Model organism / in vitro
39706863	<i>Genetic and allelic heterogeneity in 248 Indians with skeletal dysplasia</i>	Expands <i>RSPRY1</i> genotype/phenotype spectrum in a large cohort	Human clinical
33092958	<i>Substrate recognition by TRIM/TRIM-like proteins</i>	Contextualizes PRY-SPRY as a substrate-recognition module (structural analogy)	Review

Supporting TRIM/SPRY-family literature ([PMID: 36675197](#), [33989636](#), [40593126](#), [38780244](#), [37139802](#)) establishes the general biochemistry of RING + PRY/SPRY proteins as E3 ubiquitin ligases/substrate adaptors, providing an analogy-based (not SEMDFA-specific) framework for

RSPRY1 function. An endometriosis Mendelian-randomization study (PMID: 39978332) lists *RSPRY1* as a candidate eQTL target in an unrelated disease context and is not directly relevant to SEMDFA pathogenesis.

Supported hypotheses: biallelic LoF *RSPRY1* causation; autosomal-recessive inheritance with consanguinity; the progressive SEMD + short stature + facial dysmorphism + intellectual disability ± craniosynostosis phenotype; RSPRY1 action in bone-forming cells during endochondral ossification; constitutive SMAD3-dependent TGF- β /ECM dysregulation as the molecular mechanism.

Refuted / not supported: dominant inheritance, environmental/infectious causation, and gain-of-function mechanisms are not supported. Universal microcephaly (suggested by the original family) was refuted by the 2018 cohort, in which most patients were normocephalic (PMID: 30063090).

Limitations and Knowledge Gaps

1. **Very small evidence base.** ~11 patients across 5 consanguineous families underpin the entire clinical picture; per-phenotype frequencies, prognosis, and natural history are not quantifiable with confidence, and there are no epidemiologic, registry, or QoL data.
2. **Mechanism from a single functional study in fibroblasts.** The TGF- β /SMAD3 model rests primarily on one report using patient fibroblasts and knockout cell lines (PMID: 39940902); it has not been validated in chondrocytes/osteoblasts or in vivo, the cell types most relevant to the skeletal phenotype.
3. **Molecular link untested.** The direct biochemical connection between RSPRY1 (RING/SPRY, secreted) and TGF- β /SMAD3 restraint — e.g., a specific ubiquitination substrate — is inferred by analogy to TRIM proteins, not demonstrated for RSPRY1.
4. **Discordant localization annotations.** UniProt annotates RSPRY1 as "Secreted," yet a RING/SPRY E3-ligase function and the intracellular constitutive TGF- β phenotype imply intracellular activity; this apparent discrepancy is unresolved.
5. **No animal model of the disease.** No *Rspry1*-null mouse skeletal phenotype has been reported, limiting causal in-vivo confirmation and preclinical therapy testing.
6. **No therapeutics.** TGF- β inhibition is a hypothesis only; efficacy and safety in SEMDFA are untested and no trials are registered.

7. **Epidemiology absent.** No prevalence/incidence estimates, carrier frequencies, or geographic-variant maps beyond consanguineous founder observations.

Proposed Follow-up Experiments / Actions

1. **Generate an *Rspry1* mouse model** (global and cartilage-specific conditional knockout, e.g., *Col2a1-Cre*) and characterize growth-plate architecture, endochondral ossification, and craniofacial sutures to test skeletal recapitulation.
 2. **Validate the TGF- β /SMAD3 mechanism in disease-relevant cells** — patient-derived iPSC chondrocytes/osteoblasts and growth-plate organoids — with phospho-SMAD3 quantification and rescue by SMAD3 knockdown or ALK5 (TGF β R1) inhibitors (e.g., SB-431542, galunisertib).
 3. **Define the RSPRY1 interactome and substrate(s).** Use affinity purification–mass spectrometry and ubiquitination assays to test whether the RING/SPRY domain targets a TGF- β pathway component for ubiquitination, bridging the structural annotation (Finding 5) and functional mechanism (Finding 3).
 4. **Resolve subcellular localization** with endogenous tagging/immunofluorescence and secretome analysis to reconcile the "secreted" annotation with an intracellular E3-ligase function.
 5. **Preclinical therapeutic proof-of-concept:** test TGF- β /SMAD3 pathway inhibitors in the mouse model and in patient chondrocyte organoids for correction of ECM and differentiation defects.
 6. **Establish an international patient registry** to collect standardized skeletal, developmental, and pediatric HRQoL outcomes and to define natural history, penetrance, and expressivity.
 7. **Systematic variant curation** with functional assays (minigene splicing for splice variants; NMD/expression assays for truncating alleles; overexpression rescue for missense) to reclassify VUS and the conflicting splice-region ClinVar entries.
 8. **Carrier-frequency and founder-haplotype studies** in consanguineous populations (Saudi, Turkish, Indian, Peruvian) to inform targeted carrier screening and counseling.
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Report compiled from an autonomous multi-iteration literature and database investigation (7 confirmed findings; 15 papers reviewed). Evidence types are distinguished as human clinical, in vitro/patient-cell, model organism, and computational throughout.

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